

GEOGRAPHY

INTRODUCTION

Geography is about determining the significance of “place” as it relates to the natural environment, the human environment, and interactions within and between them. To investigate geographic issues, students must analyse the influences and interrelationships that give a place its distinctive characteristics and thus its spatial importance. Geographic analysis also requires an investigation of the economic, environmental, social, and political perspectives that relate to an issue. The application of the concepts of geographic thinking, spatial skills, and the use of field studies are central to the geographic inquiry process and the learning of geography.

Strands

The Grade 11 and 12 geography courses each have five strands. In all of these courses, strand A, Geographic Inquiry and Skill Development, is followed by four content strands, which are organized thematically.

Citizenship Education

The expectations in the Grade 11 and 12 geography courses provide opportunities for students to explore a number of concepts connected to the citizenship education framework (see page 13).

The Concepts of Geographic Thinking

The four concepts of geographic thinking – spatial significance, patterns and trends, interrelationships, and geographic perspective – underpin thinking and learning in all geography courses in the Canadian and world studies program. At least one concept of geographic thinking is identified as the focus for each overall expectation in the content strands of these courses. The following chart describes each concept and provides sample questions related to it. These questions highlight opportunities for students to apply a specific concept in their studies. (See page 16 for a fuller discussion of the concepts of disciplinary thinking.)

Spatial Significance
This concept requires students to determine the importance of a place or region. They explore the connections that exist between the geographical location and physical characteristics of a site and analyse the unique relationships that exist in and between the natural and human environments in a particular place. Students come to understand that the significance of the same place may be different for humans, animals, and plants. Related Questions* – What does a population settlement map tell us about the preferred range of latitude for settlement in Canada? (CGC1P, B2.2) – How does the official plan for your community address urban sprawl? (CGC1D, E2.3) – How might the uniqueness of a destination make it more appealing to some and less appealing to others? (CGG3O, B1.2) – How does the location of a site influence the feasibility of different types of power generation there? (CGR4E, E3.4) – What are some of the challenges that communities in permafrost regions face in providing safe drinking water and disposing of waste? (CGR4M, E2.3)
Patterns and Trends
This concept requires students to recognize characteristics that are similar and that repeat themselves in a natural or human environment (patterns) and characteristics or traits that exhibit a consistent tendency in a particular setting over a period of time (trends). The characteristics may be spatial, social, economic, physical, or environmental. Students analyse connections between characteristics to determine patterns; they analyse connections between those characteristics over time to determine trends. Related Questions – What trends do you see in the use of alternative energy in Ontario? (CGC1P, A1.5) – What pattern or patterns do you see in the location of First Nations reserves across Canada? (CGC1D, D3.1) – What factors have led to population movement in the region? What specific locations have attracted migrants? Why? (CGD3M, E2.1) – In what ways has technological change amplified the impact of population growth on the natural environment? In what ways has it diminished it? (CGW4U, C2.2)

* These questions are drawn directly from the overview charts that precede the geography courses and from the sample questions that accompany many specific expectations. To highlight the continuity between the geography courses in Grade 11 and 12 and those in Grade 9, and to show possible development in the use of the concepts of geographic thinking over those grades, the chart includes some questions from the Grade 9 geography curriculum as well.

Interrelationships

This concept requires students to explore connections within and between natural and human environments. The interconnected parts of an environment or environments work together to form a system. Students must understand the relationships that exist within a system and then critically analyse the relationships between systems in order to determine the impact they have on one another.

Related Questions

- How does surrounding farmland support a community, and what stresses might the community place on the farmland? (CGC1P, E2.1)
- How might the breaking up of continental ice in Greenland and the Antarctic affect Canada's coastline? (CGC1D, B2.1)
- Why do we need to understand the earth's physical processes in order to predict their potential hazards? (CGF3M, Overview)
- How can international trade policies and corporate actions influence disparities within or between ecumenes? (CGU4M, D1.2)

Geographic Perspective

This concept requires students to consider the environmental, economic, political, and/or social implications of the issues, events, developments, and/or phenomena that they are analysing. In order to solve problems, make decisions or judgements, or formulate plans of action effectively, students need to develop their ability to examine issues from multiple perspectives.

Related Questions

- In what ways can cultural diversity enrich the life of a community? (CGC1P, D1.2)
- Does the financial benefit of extracting natural resources justify related social and/or environmental impacts? (CGC1D, A1.6)
- How might spatial technologies help governments assess the economic, environmental, infrastructural, and social risks of a potential disaster? Why is it important for governments to know and understand these risks? (CGT3O, D1.2)
- Why might some people believe that there is an ethical imperative to share the world's resources equitably? (CGW4C, Overview)
- Why might military and/or government satellite information be shared with or withheld from certain countries or military alliances? What are the implications of doing so? (CGO4M, D2.1)

The Geographic Inquiry Process

In each of the geography courses in the Canadian and world studies curriculum, strand A focuses explicitly on the geographic inquiry process, guiding students in their investigations of issues, events, developments, and/or various geographic phenomena. This process is *not* intended to be applied in a linear manner: students will use the applicable components of the process in the order most appropriate for them and for the task at hand. Although strand A covers all of the components of the inquiry process, it is important to note that students apply skills associated with the inquiry process throughout the content strands in each course. (See page 32 for a fuller discussion of the inquiry process in the Canadian and world studies program.)

The following chart identifies ways in which students may approach each of the components of the geographic inquiry process.

Formulate Questions
Students formulate questions: – to explore various events, developments, issues, and/or phenomena that are related to the overall expectations in order to identify the focus of their inquiry – to help them determine which key concept or concepts of geographic thinking are relevant to their inquiry – that reflect the selected concept(s) of geographic thinking – to develop criteria that they will use in evaluating data, evidence, and/or information; in making judgements, decisions, or predictions; in reaching conclusions; and/or in solving problems
Gather and Organize
Students: – collect relevant qualitative and quantitative data, evidence, and information from field studies^a and a variety of primary and secondary sources,^b including visuals^c and community resources^d – determine if their sources are credible, accurate, and reliable – identify the purpose and intent of each source – identify the points of view in the sources they have gathered – use a variety of methods to organize the data, evidence, and/or information they have gathered – record the sources of the data, evidence, and information they are using – decide whether they have collected enough data, evidence, and/or information for their inquiry

a. Field studies may include, but are not limited to, studies in local neighbourhoods, school grounds, and various sites that allow students to explore different land uses (e.g., recreational, commercial, industrial, and transportation uses).

b. Primary sources may include, but are not limited to, census data, land claims, letters, photographs, speeches, and works of art. Secondary sources may include, but are not limited to, documentaries and other films, news articles, reference books, and most websites.

c. Visuals may include, but are not limited to, satellite images, maps, globes, models, graphs, and diagrams.

d. Community resources may include, but are not limited to, local conservation areas, resources from community groups and associations, government resources, and local plans.

Interpret and Analyse
Students: – analyse data, evidence, and information, applying the relevant concepts of geographic thinking (see preceding chart) – use different types of graphic organizers to help them interpret and/or analyse their data, evidence, and information – identify the key points or ideas in each source – analyse graphs, charts, diagrams, and maps – construct graphs, charts, diagrams, and maps to help them analyse the issue, event, development, or phenomenon they are investigating – analyse their sources to determine the importance of an issue, event, development, or phenomenon for individuals or groups – identify biases in individual sources – determine if all points of view are represented in the source materials as a whole, and which, if any, are missing
Evaluate and Draw Conclusions
Students: – synthesize data, evidence, and information, and make informed, critical judgements based on that data, evidence, and information – determine the short- and long-term impact of an event, development, issue, or phenomenon on people and/or places – reach conclusions about the subject of their inquiry, and support them with their data, evidence, and information – make predictions based on their data, evidence, and information – determine the ethical implications of an issue or action – determine the action required, where appropriate
Communicate
Students: – use appropriate forms (e.g., oral, visual, written, multimedia) for different audiences and purposes – communicate their arguments, conclusions, predictions, and plans of action clearly and logically – use geographical terminology and concepts correctly and effectively – cite sources, using appropriate forms of documentation

Living in a Sustainable World, Grade 12

Workplace Preparation

CGR4E

This course examines the impact of human activity on the natural environment. Students will explore the use of natural spaces and resources and the effects of planning decisions and consumer choices on natural systems. Students will apply the concepts of geographic thinking and the geographic inquiry process, including spatial technologies, to investigate practical solutions to environmental issues, enabling them to make more sustainable decisions at home, in the workplace, and in the local community.

Prerequisite: Issues in Canadian Geography, Grade 9, Academic or Applied

OVERVIEW

The course has five strands. Instruction and learning related to the expectations in strand A are to be interwoven with instruction and learning related to expectations from the other four strands. Strand A must not be seen as independent of the other strands. Student achievement of the expectations in strand A is to be assessed and evaluated *throughout* the course.

Strand A

A. Geographic Inquiry and Skill Development
Overall Expectations
A1. Geographic Inquiry: use the geographic inquiry process and the concepts of geographic thinking when investigating issues relating to the natural environment and sustainability
A2. Developing Transferable Skills: apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset

(continued)

Overview *(continued)*

Throughout this course, when planning instruction, teachers should weave the expectations from strand A in with the expectations from strands B–E.

Strands B–E

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
B. Species and Spaces		
B1. Endangered Species and Spaces: identify species and natural places at risk in different parts of the world, and compare approaches that various countries have taken for their protection (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	It is important for countries to have strategies to protect endangered species and spaces.	What are some of the ways in which countries work together to protect a species that crosses international borders?
B2. Human Impacts on Ecosystems: explain how human settlement and activities alter ecosystems (FOCUS ON: <i>Patterns and Trends; Geographic Perspective</i>)	Humans have had and continue to have a direct impact on the natural environment.	What stands in the way of cooperation on such issues?
B3. Ecosystem Characteristics: describe the characteristics of different types of ecosystems, and explain their relationships with natural processes in the Earth system (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	The earth and its ecosystems are made up of many interacting components.	What do you think is the most significant impact that humans have had on an ecosystem? How are the various parts of an ecosystem related to each other? What might happen if one element is changed or no longer present?
C. Sustainability of Natural Resources		
C1. Strategies and Stewardship Initiatives: assess the contributions of stewardship initiatives by groups and individuals to the sustainable use and management of natural resources, locally, nationally, and globally (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	Many individuals, groups, and countries around the world are practising environmental stewardship.	How do your personal choices and behaviours affect the use of natural resources?
C2. Resource Development and Impacts: analyse impacts and issues related to the development and use of natural resources (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	The ways in which people extract or harvest natural resources and use them can have social, economic, political, and environmental impacts.	How do competing interests and ideas affect the extraction or harvesting of a natural resource and the way it is used?
C3. Distribution and Use of Natural Resources: analyse patterns and trends in the availability and use of natural resources (FOCUS ON: <i>Spatial Significance; Patterns and Trends</i>)	The location of a resource can determine how it is extracted or harvested and how it is used.	How might the loss of a natural resource affect a community? Why does the process used to extract a natural resource depend on where the resource is located?

* See page 17 for a discussion of the purpose of big ideas and framing questions.

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
D. Ecosystems and Human Activity		
D1. Protecting the Natural Environment: assess the role of various strategies, organizations, and agreements in reducing the impact of human activity on the environment (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	Individuals, groups, and countries have developed strategies to lessen the impact of humans on the environment.	Do all organizations and countries agree on how to protect the environment? Why might there be conflicting strategies for reducing the human impact on the environment?
D2. Impacts of Human Activities: analyse impacts of human activities on ecological processes and on plant and animal species (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Human activity can alter and harm natural spaces and species.	Why might an individual, a company, or a country introduce a non-native species or a pollutant into a local environment?
D3. Ecological Processes: explain how various ecological and biological processes sustain life on Earth (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	The earth sustains life.	How does life on Earth rely on the earth's physical processes?
E. Community Action		
E1. Working Together: assess the contribution of various individual, workplace, and community initiatives to reducing the human impact on the natural environment (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	People need to work together to lessen the impact of humans on the environment.	What are some local strategies and policies that have been proposed or adopted to protect the environment? Why might people not agree about the purpose or value of these strategies or policies?
E2. Ecological Footprints: assess impacts of human behaviour on the natural environment (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	The behaviour and choices of individuals can have a significant impact on the environment.	How might you reduce your ecological footprint?
E3. Community Infrastructure: assess environmental impacts of various types of infrastructure, systems, and services at the community and regional levels, and assess ways of reducing these impacts (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Local practices can have important impacts on the natural environment.	How is infrastructure in your community being changed to lessen its impact on the natural environment? What support is there for these changes?

A. GEOGRAPHIC INQUIRY AND SKILL DEVELOPMENT

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1. Geographic Inquiry:** use the geographic inquiry process and the concepts of geographic thinking when investigating issues relating to the natural environment and sustainability;
- A2. Developing Transferable Skills:** apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset.

SPECIFIC EXPECTATIONS

A1. Geographic Inquiry

Throughout this course, students will:

- A1.1** formulate different types of questions to guide investigations into issues affecting the natural environment (e.g., factual questions: *Which three countries in the world have the greatest freshwater reserves?* comparative questions: *How does the ecological footprint of the average Dane or German compare with that of the average Canadian?*; causal questions: *How does your choice of consumer goods affect the consumption of natural resources?*)

- A1.2** select and organize relevant data and information on geographic issues from a variety of primary and secondary sources (e.g., primary: *raw data from fieldwork, both quantitative and qualitative; photographs; satellite images*; secondary: *published statistics, newspapers, books, atlases, geographic magazines, websites, graphs, charts, digital and print maps*), ensuring that their sources represent a diverse range of perspectives

Sample questions: “How would you use photographs of your community from different time periods to determine how much the amount of developed land has grown? Where might you find these photographs and related information?” “What type of data and information do you need to collect in order to analyse the impact that a new park has had on your community?” “What types of maps and graphs will help you analyse air quality in your region?”

- A1.3** assess the credibility of sources and information relevant to their investigations (e.g., by

considering how the data are constructed to support the author’s point of view, the possible bias of the author, the expertise of the author, the accuracy of the text and supporting data, the intended audience, the purpose of the messaging, the context in which the information was presented)

Sample questions: “Who are the authors of this source, and what are their qualifications for writing on this topic? What organizations are the authors affiliated with? Do these affiliations suggest a bias? What are the objectives of the organization that sponsored or published their work?” “Have you consulted other sources that represent other points of view?” “Which source is most credible and why?”

- A1.4** interpret and analyse data and information relevant to their investigations, using various tools, strategies, and approaches appropriate for geographic inquiry (e.g., *interpret diagrams illustrating the environmental impacts of a product throughout its life cycle, from resource extraction to disposal; identify trends and correlations by analysing graphs and charts showing the amounts and types of solid waste generated by households, institutions, and businesses in the community; use decision-making templates to analyse points of view on an issue; use graphic organizers to outline various perspectives on natural resource use*)

Sample questions: “What type of graphic organizer would you use to help analyse the costs and benefits of separating compostable material from garbage that is to be sent to a landfill?” “What type of information might you include in geographic information system (GIS) data layers in order to analyse the global distribution of pollutants emitted by vehicles that use fossil fuels?”

A1.5 use the concepts of geographic thinking (i.e., spatial significance, patterns and trends, interrelationships, geographic perspective) when analysing and evaluating data and information, formulating conclusions, and making judgements about issues affecting the natural environment (e.g., *use the concept of spatial significance to analyse the global distribution of fresh water; use the concept of patterns and trends to identify opportunities for using alternative transportation methods; use the concept of interrelationships to interpret the contributions of various natural and human factors to climate change; use the concept of geographic perspective to analyse the social, political, economic, and environmental significance of an environmental event such as Earth Hour*)

Sample questions: “How might the concept of spatial significance help you explain the importance of fresh water as a factor in the location of population settlements?” “How might an understanding of patterns and trends help you analyse the effects on waste disposal of using reusable cloth bags for grocery shopping?” “How might an understanding of interrelationships guide your examination of the connections between the time when you use electricity and the cost to produce the electricity?” “How can geographic perspective help you analyse the significance of Aboriginal peoples’ beliefs about the natural environment?”

A1.6 evaluate and synthesize their findings to formulate conclusions and/or make informed judgements or predictions about the issues they are investigating

Sample questions: “What did you find out about the impacts of global settlement trends on various ecosystems? What strategies do you think need to be implemented to respond to these impacts?” “What did you find out about the relationship between the processes used to extract or harvest a natural resource and the location of that resource? Why is this relationship an issue?”

A1.7 communicate their ideas, arguments, and conclusions using various formats and styles, as appropriate for the audience and purpose (e.g., *a debate for classmates on the use of water; a video for a Grade 9 class showing different points of view about the environmental impacts of our food choices; a webcast or podcast for the general public on strategies for managing various types of waste; a photo essay for a local community group to illustrate the importance of saving community park space; a public service announcement to illustrate ways of reducing one’s carbon footprint*)

Sample questions: “What kind of information does your audience need? In how much detail? What format and approach would be most

effective in conveying your information to this particular audience?”

A1.8 use accepted forms of documentation (e.g., *footnotes, author/date citations, reference lists, bibliographies, annotated bibliographies, credits*) to reference different types of sources (e.g., *websites, blogs, books, articles, films, data*)

A1.9 use appropriate terminology when communicating the results of their investigations (e.g., *vocabulary specific to their inquiry; terminology related to geography and to the concepts of geographic thinking*)

A2. Developing Transferable Skills

Throughout this course, students will:

A2.1 describe ways in which geographic investigation can help them develop skills, including spatial skills and the essential skills in the Ontario Skills Passport (e.g., *reading graphic texts, writing, graphing, computer use, use of spatial technologies, oral communication, numeracy, decision making, planning, management, finding information, problem solving*), that can be transferred to the world of work and to everyday life

A2.2 apply in everyday contexts skills and work habits developed through geographic investigation (e.g., *use critical thinking, mapping, and graphing skills to deepen their understanding of a global environmental issue; use listening skills to consider multiple perspectives when discussing an issue; use spatial skills to map the distribution of a particular plant or animal species; apply work habits such as collaboration when working with a team in the community to influence a local planning decision*)

A2.3 apply the concepts of geographic thinking when analysing current events involving geographic issues (e.g., *use the concept of spatial significance to analyse the spread of a disease carried by an insect; use the concept of patterns and trends to analyse theories about the decline of honeybees and other pollinators; use the concept of interrelationships to analyse connections between agricultural issues and climate change; use the concept of geographic perspective to analyse the potential impact of buying locally*) in order to enhance their understanding of these issues and their role as informed citizens

A2.4 identify some careers in which a geography background might be an asset (e.g., *garden and landscape specialist, vehicle operator, forestry worker, fisheries worker, real estate agent, park or conservation area employee, groundskeeper*)

B. SPECIES AND SPACES

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1. Endangered Species and Spaces:** identify species and natural places at risk in different parts of the world, and compare approaches that various countries have taken for their protection (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- B2. Human Impacts on Ecosystems:** explain how human settlement and activities alter ecosystems (**FOCUS ON:** *Patterns and Trends; Geographic Perspective*)
- B3. Ecosystem Characteristics:** describe the characteristics of different types of ecosystems, and explain their relationships with the natural processes in the Earth system (**FOCUS ON:** *Spatial Significance; Interrelationships*)

SPECIFIC EXPECTATIONS

B1. Endangered Species and Spaces

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

- B1.1** assess criteria for classifying species or natural spaces at risk, and identify species and spaces that are at risk in different regions of the world
Sample questions: “What criteria determine whether a space or species is at risk?” “Who decides whether a species can be designated as being at risk?” “Are there any species at risk in your area?” “Should we protect some species more than others?”
Using spatial skills: Students can illustrate the global distribution of vulnerable species and spaces by annotating a base map of the world’s major ecosystems with information about species and places currently at risk in these ecosystems and the sustainability challenges that they face.
- B1.2** explain why selected natural spaces and species are at risk
Sample questions: “What are some of the more common threats to the survival of species and natural spaces?” “What are the major threats to the species and spaces that you are investigating? How have these threats developed?” “What do you think is the biggest threat to species and natural places today? What criteria would you use to help you decide?”

- B1.3** compare strategies that various countries have used to protect natural habitats and plant and wildlife species (*e.g., the national park systems in Canada and the United States; Sweden’s sixteen environmental quality objectives versus Australia’s Biodiversity Conservation Strategy as ways of achieving a broad set of environmental protection goals; processes used in different countries to designate species at risk or set harvesting limits*)

Sample questions: “How does the protection or preservation of spaces help support the preservation of species?” “What is the difference between preservation and conservation?” “How does the work of non-governmental conservation organizations, such as the Nature Conservancy of Canada or Britain’s National Trust, complement the role of national parks systems in protecting ecosystems?”

B2. Human Impacts on Ecosystems

FOCUS ON: *Patterns and Trends; Geographic Perspective*

By the end of this course, students will:

- B2.1** describe the beliefs of various groups of people, including indigenous peoples, about the natural environment, and explain how these beliefs have contributed to the preservation or loss of natural spaces
Sample questions: “How do the art and stories of various Aboriginal groups reflect their views

of the natural environment?" "How do various resource-based industries view their responsibilities for protecting species and natural spaces?" "What arguments do environmental non-governmental organizations use to promote the protection of the natural environment?"

B2.2 assess impacts of human settlement and activities on selected ecosystems (*e.g., deforestation and loss of habitat from resource development in the boreal forest; loss of habitat and degradation of water quality from aggregate mining and increased agriculture in the Carolinian forest in southern Ontario; water pollution from industry, agriculture, and human waste in the Florida wetlands; threats to coral reefs from rising water temperatures and increased human interaction*)

Sample questions: "What are the benefits of protecting the boreal forest, coral reefs, or wetlands?" "Can population growth and preservation of natural environments coexist, or does there have to be a winner and a loser?"

Using spatial skills: Using a world base map, students can add layers showing ecosystem and human settlement data in order to identify and analyse ecosystems that have been affected by population growth. Web diagrams can be created to illustrate how an impact on one element of an ecosystem can affect other elements of the ecosystem.

B2.3 assess impacts of population growth on the natural environment in their local area, and identify some of the social, economic, and political consequences of these impacts (*e.g., social: the loss of aesthetic qualities and recreational areas as a consequence of loss of green space; economic: the cost of building infrastructure to manage stormwater runoff and flooding as a consequence of an increase in paved area and loss of wetlands; political: the increased need for government to monitor and regulate air and water quality as a consequence of a greater number and variety of sources of pollution*)

Sample questions: "How has population growth affected local streams and wetlands? What has been the effect on fish and wildlife? Has the community invested in measures to protect water quality in the area?" "Has the community become more car-dependent? How has that shift affected air quality? What is the community doing to control air pollution and its effects?" "Have new industries come into the area? How have they affected population growth, the economy, and the local environment?" "Has the quality of life changed in the community as it has become more detached

from its natural surroundings?" "Have there been disagreements in the community about whether it should grow or how it should grow?"

B3. Ecosystem Characteristics

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

B3.1 explain ways in which living things depend on other components of the Earth system (*e.g., atmosphere, lithosphere, hydrosphere*) and describe some of the interactions between them

Sample questions: "What does the term 'global commons' mean? Which components of the global commons are essential to the existence of life on Earth?" "How does the biosphere affect the atmosphere?"

Using spatial skills: Students can use a flow diagram with visuals to illustrate interrelationships between different components of the Earth system.

B3.2 identify the elements of an ecosystem and explain how they interact

Sample questions: "How do climate and soils affect the types of vegetation in an ecosystem? How does the vegetation affect the wildlife?" "Of all the elements in an ecosystem, which one has the greatest effect on the other elements and on the characteristics of that ecosystem?" "What is the place of humans in an ecosystem? In what ways do they depend on the other components of the ecosystem? What are some of the ways in which they affect other components of the ecosystem?"

B3.3 describe the characteristics of different types of ecosystems in different regions of the world

Sample questions: "What are the similarities and differences between the Siberian tundra and the Sahara desert?" "How is the Brazilian rainforest different from the rainforest found along the west coast of North America?" "Why are some wildlife species, such as the polar bear or the koala bear, found only in certain regions of the world?"

Using spatial skills: Students can support their investigations of different kinds of ecosystems by adding an ecosystem layer to a world base map and adding annotations, including visuals, to describe the identifying characteristics of particular ecological regions.

C. SUSTAINABILITY OF NATURAL RESOURCES

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1. Strategies and Stewardship Initiatives:** assess the contributions of stewardship initiatives by groups and individuals to the sustainable use and management of natural resources, locally, nationally, and globally (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- C2. Resource Development and Impacts:** analyse impacts and issues related to the development and use of natural resources (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- C3. Distribution and Use of Natural Resources:** analyse patterns and trends in the availability and use of natural resources (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)

SPECIFIC EXPECTATIONS

C1. Strategies and Stewardship Initiatives

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- C1.1** analyse selected responses by governments, industries, and/or non-governmental organizations to resource-related environmental concerns (*e.g., installing stack scrubbers to reduce emissions that cause smog and acid rain; banning the use of some chemicals; passing laws and regulations to control discharges of pollutants into water bodies; rehabilitating tailing ponds and open-pit mines to restore habitat; using sustained-yield practices to maintain healthy fish stocks and forests; creating public awareness of an environmental issue and building support for action*), and assess the contribution of these actions to responsible and sustainable resource use and management

Sample questions: “What was the problem that led to this response?” “Was government regulation needed to support the response? Did non-governmental organizations play a role by influencing public opinion? How did the industry respond to the problem?” “What benefits did this response lead to, for the environment, the public, and the industry?”

- C1.2** describe ways in which individuals can contribute to the sustainable use of natural resources (*e.g., by buying, selling, or donating good-quality used products so that they may be*

reused; reducing personal consumption; recycling; buying fair-trade products; supporting environmental non-governmental organizations)

Sample questions: “How does your consumer behaviour affect the consumption of natural resources? How would the demand for natural resources be affected if people bought only what they needed instead of what they wanted?” “What effect does your choice of transportation have on the use of natural resources?”

C2. Resource Development and Impacts

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- C2.1** assess the environmental impact of a particular consumer product (*e.g., cellphone, bicycle, car, golf club*) over its life cycle, from the extraction of the resources needed to produce it, through its production and use, to its disposal

Sample questions: “At what stage in its life cycle does your product have the greatest environmental impacts?” “Why do you have to pay a disposal fee when you buy some products?”

Using spatial skills: Students can create flow diagrams or concept webs to illustrate the steps and the impacts involved in various stages of the life cycle of a product.

C2.2 analyse the perspectives of stakeholders concerned with a resource management or development issue

Sample questions: “How do various stakeholders, such as First Nations and Inuit people, local residents and businesses, governments, oil companies, and environmental organizations, perceive the Alberta oil sands development? What arguments do they use to support their positions? Which arguments do you find most persuasive? Why?” “How might different points of view about the development of a resource, such as oil or diamonds, influence the choices you make as a consumer?”

C3. Distribution and Use of Natural Resources

FOCUS ON: *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

C3.1 analyse factors affecting the global distribution of major types of renewable and non-renewable natural resources, and assess some of the consequences of the distribution and availability of these resources

Sample questions: “What factors determine where fresh water can be found? Why is fresh water not available in desert areas?” “In what parts of the world are metallic and non-metallic mineral deposits and fossil-fuel deposits found?” “Why might a country possess a natural resource but not extract it?” “How might the availability of a resource or the lack thereof in a particular country influence the way of life of people in that country?” “Why might countries go to war over natural resources?” “Where in Canada are the best places for the production of wind, solar, geothermal, and tidal energy?”

Using spatial skills: Students can select a resource, such as fresh water, oil and gas, bauxite, chromite, coltan, or arable land, and construct maps showing where this resource is located across the world. Annotations can provide further details about the resource, such as its abundance, quality, and accessibility. Teachers should ensure that a broad range of resources is represented in the class. Students can then compare their maps to determine which countries of the world are resource rich and which ones appear to be resource poor.

C3.2 analyse the consumption of resources in countries at different levels of development (*e.g., per capita consumption of oil and gas, steel, fresh water, grain, and fertilizer in selected developed, developing, and underdeveloped countries*)

Sample questions: “Why might different countries use different amounts of oil and gas?” “How do the consumption statistics correlate with GDP per capita in each country? Are there factors other than national wealth that might explain why some countries use more or less of a resource than other countries? Is a country’s consumption of a particular resource affected by the availability of the resource in that country or the availability of alternative resources?” “Why do we use per capita figures for these comparisons instead of a country’s total consumption?” “How have China’s imports of resources changed as the country has become more prosperous?” “How does water use differ in developed and underdeveloped countries? What factors influence water use?”

C3.3 explain changes in the importance of a variety of resources over time (*e.g., feather quills [once used as pens], kerosene [once used as lamp oil], coal [once used for heat and steam power, now used for electricity], lithium [now used in rechargeable batteries]*)

Sample questions: “How is a resource defined?” “Are some resources more important than others?” “Can you identify some things that were considered resources in the recent and distant past that are not considered resources today? Can you identify some things that are considered resources in one culture but not another? What are some resources that we use today that might not be resources in the future?”

C3.4 describe how population growth affects the availability and consumption of a variety of resources

Sample questions: “How does population growth affect the consumption of oil?” “Does population growth increase the need to find more distant sources of supply for some resources, such as food and water?” “Can population growth lead to the loss of some resources?” “Why might it be important to preserve agricultural land close to urban communities?”

Using spatial skills: Students can use graphs and maps to illustrate the link between urban growth and the loss of arable land in Ontario.

C3.5 analyse factors that might influence people to use alternatives to certain natural resources or find alternative sources of supply (*e.g., scarcity of a resource, political uncertainty, environmental consequences, ethical concerns*)

Sample questions: “Why are we increasingly dependent on expensive and possibly harmful techniques, such as fracking or offshore drilling, for our oil and gas supplies?” “Why are some

companies considering the possibility of extracting metals from the ocean floor or from asteroids and nearby planets?” “What are the factors driving the demand to replace fossil fuels with alternative sources of energy?” “What are conflict minerals, and how did ethical concerns about them support the development of diamond mining in Canada?”

D. ECOSYSTEMS AND HUMAN ACTIVITY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1. Protecting the Natural Environment:** assess the role of various strategies, organizations, and agreements in reducing the impact of human activity on the environment (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- D2. Impacts of Human Activities:** analyse impacts of human activities on ecological processes and on plant and animal species (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- D3. Ecological Processes:** explain how various ecological and biological processes sustain life on Earth (**FOCUS ON:** *Patterns and Trends; Interrelationships*)

SPECIFIC EXPECTATIONS

D1. Protecting the Natural Environment

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- D1.1** describe strategies that have been used to reduce pollution from human activities (*e.g., banning the use of polluting chemicals; reformulating products to eliminate harmful ingredients, such as phosphates in detergents; installing filtration devices, such as stack scrubbers or catalytic converters, to remove pollutants from emissions; using non-chemical alternatives to aerosol sprays, pesticides, and other products containing harmful chemicals*)

Sample questions: “How does scientific research contribute to the discovery and solution of pollution problems? What part do governments, universities, and private industry play in this research?” “How can individuals help to minimize pollution?”

- D1.2** assess ways in which international organizations and agreements help to protect the global environment (*e.g., organizations: Greenpeace, World Wildlife Fund, Sierra Club, Intergovernmental Panel on Climate Change, United Nations Environment Programme; agreements: Kyoto Protocol, Montreal Protocol, Great Lakes Water Quality Agreement*)

- D1.3** assess physical and biological changes that have occurred in a local area (*e.g., a creek, a*

wetland, a woodlot, a quarry, a mine) as a result of a rehabilitation or restoration project

Using spatial skills: Students can support their investigation of environmental remediation by conducting a field study of a local area that has undergone restoration or rehabilitation.

D2. Impacts of Human Activities

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

- D2.1** describe ways in which humans have altered ecological processes (*e.g., by draining wetlands, paving over land, contaminating land and water with chemicals, deforestation, overharvesting*), and explain the impacts of these activities on natural systems (*e.g., loss of habitat, lowering of the water table, increase in levels of carbon dioxide in the atmosphere, loss of species, disruption of food chains, formation of urban heat islands and related impacts on air and water quality*)

- D2.2** assess the effects on a natural system of the removal or introduction of a species (*e.g., rabbits in Australia, zebra mussels in the Great Lakes, apple snails in Hawaii, Monterey pines in New Zealand, brown tree snakes on Guam*)

Sample questions: “Why would people deliberately introduce a foreign species into a natural environment?” “Why do customs officers ask if you are bringing any plants or animals into the country?”

D2.3 identify major types of pollutants and their sources (e.g., *toxic chemicals and heavy metals, smog, acid precipitation, bacteria in water, nutrients in water, noise, light*), and explain their effects on plants and animals, including humans (e.g., *reduced fertility, cancers, birth defects, cardiorespiratory disorders, neural disorders, detrimental confusion among sea turtle hatchlings*)

Sample questions: “How is it possible for large fish and the predators that eat them to have large amounts of toxic pollutants in their tissues when only a small quantity of these pollutants is present in the water?” “How do agricultural fertilizers or phosphates in detergents affect plant and animal life in lakes?” “Why is it sometimes difficult to prove that a particular pollutant has a specific effect? Why might this make it difficult for people to understand or believe that a pollution problem exists?” “Why might there be many points of view about the effects of pollution?”

Using spatial skills: Constructing annotated maps showing pollution impacts and pollution sources for various regions can help students in their investigation of distribution patterns and relationships between affected areas and sources. Graphs can be added to show trends in pollutant emissions and environmental concentrations over time, as well as trends in impacts.

Sample questions: “How do the non-living components of an ecosystem affect the type and amount of vegetation in the system?” “What determines the number and variety of animals in an ecosystem?” “How do decomposers support the sustainability of an ecosystem?”

D3.2 describe key nutrient cycles and energy flows in the Earth system (e.g., *the carbon/oxygen, nitrogen, and hydrologic cycles*)

Sample questions: “Where do plants get the energy they need to grow?” “What happens to the energy stored in uneaten food and in dead plants and animals?” “How does lightning help to support life on Earth?” “Why is it important to understand the hydrologic cycle?”

Using spatial skills: Students can create flow diagrams to illustrate the processes involved in major nutrient cycles, such as the water, carbon, and nitrogen cycles.

D3.3 describe the process of natural ecological succession in a variety of natural communities

Sample questions: “What changes happen to a forest over time?” “How does a beach change naturally over time?” “How does a volcanic island become repopulated with plants and animals after an eruption?”

D3. Ecological Processes

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

D3.1 describe relationships between the non-living and living components of an ecosystem (e.g., non-living: *sunlight, temperature, precipitation, soils*; living: *producers, consumers, and decomposers*)

E. COMMUNITY ACTION

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1. Working Together:** assess the contribution of various individual, workplace, and community initiatives to reducing the human impact on the natural environment (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- E2. Ecological Footprints:** assess impacts of human behaviour on the natural environment (**FOCUS ON:** *Patterns and Trends; Interrelationships*)
- E3. Community Infrastructure:** assess environmental impacts of various types of infrastructure, systems, and services at the community and regional levels, and assess ways of reducing these impacts (**FOCUS ON:** *Spatial Significance; Interrelationships*)

SPECIFIC EXPECTATIONS

E1. Working Together

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- E1.1** analyse the contributions of selected individuals, businesses, and organizations to resolving environmental issues of concern to their local community

Sample questions: “What have individuals, such as David Suzuki, Robert Kennedy Jr., and Maude Barlow done to raise awareness about environmental concerns or promote solutions to environmental problems that are of concern to your community?” “What do some businesses that operate in your community do to reduce the environmental impact of their operations or of the products they sell? Do any businesses in your community sell fair-trade products? What environmental criteria do products have to meet in order to receive fair-trade designation?” “Are there any organizations in your community that deal with local environmental concerns?” “How effective was this local initiative? What criteria would you use to judge its effectiveness?”

- E1.2** evaluate the effectiveness of various public awareness campaigns and initiatives in promoting positive environmental change (e.g., *Earth Day, Earth Hour, Flick Off, Every Kilowatt Counts, One Hundred Kilometre Diet, the Free Rice Challenge of the World Food Programme, Waste-Free Lunch Challenge, Idle No More*)

Sample questions: “How much influence do you think various awareness initiatives have had on

public opinion and on people’s behaviour?” “How influential have the media been in affecting public opinion about environmental concerns?” “What is the best way to encourage teenagers to live in a more environmentally friendly way?”

E2. Ecological Footprints

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

- E2.1** calculate their ecological footprints and create a plan to reduce personal consumption and waste

Sample questions: “How does your ecological footprint compare with the Canadian average?” “What areas of consumption had the greatest effect on your footprint? How might you reduce this part of your footprint?”

- E2.2** compare typical ecological footprints of people in countries in various parts of the world, and identify, through analysis, possible causes of the differences between them

- E2.3** analyse the environmental impact of a variety of household appliances and products (e.g., *the energy consumption of various appliances; the effects of chemicals used in cleaning agents, paint, fertilizers, pesticides*)

Sample questions: “What does the Energy Star label on an appliance tell you about its environmental impact?” “How might buying a more expensive appliance save you money?”

E2.4 analyse the impact of consumer behaviour on the environment (*e.g., transportation choices, water and energy consumption, product choices*)

Sample questions: “How can a person’s shopping habits and choices affect the sustainability of natural resources?” “How would the consumption of resources change if people bought less of what they wanted but didn’t really need?” “Do businesses have a responsibility to promote and support more sustainable consumer behaviour? How can consumers encourage businesses to reduce the environmental impact of their products?” “In what ways do businesses encourage recycling of their products?”

E3. Community Infrastructure

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

E3.1 assess the environmental impact of water supply and wastewater management systems in various communities

Sample questions: “What precautions do people need to take if their water comes from a well?” “How do older sewer systems that carry both stormwater and household sewage threaten local rivers and lakes? Are there any such systems in your community? What is being done to manage the problems associated with these systems?” “What is meant by the term *grey water*? What can it be used for? What are the environmental benefits of reusing grey water?” “Why might a disproportionate number of First Nations communities be on a long-standing boil-water advisory?”

E3.2 assess the environmental impact of various methods of waste disposal (*e.g., landfilling, incineration, recycling, composting, transporting to other communities*)

Sample questions: “Why should people not put hazardous waste, such as batteries and paint, in their garbage? What is the environmental impact of these materials when they are sent to a landfill? What kinds of waste should be considered hazardous?”

E3.3 describe the environmental impact of transportation in their community, and assess the possibility of using alternative approaches to meeting the community’s transportation needs

Sample questions: “What forms of transportation are used most in your community? What are the environmental impacts associated with their use?” “Are there alternative forms of transportation that could be used? Would most people in the community have access to them? Could the community afford to build the infrastructure that these alternative systems would need? How could people be persuaded to use these alternative systems?” “Are there ways of reducing the environmental impact of the transportation systems that are already in use?”

E3.4 describe the environmental impact of different methods of power generation, and assess the possibility of using alternative sources of energy generation (*e.g., solar, water, wind, geothermal*) to provide electricity for their own community

Sample questions: “What are the major sources of electrical power in Ontario, and what environmental impacts are associated with each of these sources?” “How does the location of a site influence the feasibility of different types of power generation there?” “What alternative sources would be practical for our area?”